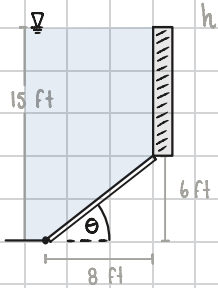
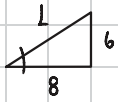
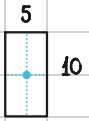
Problema 1



Compuerta : 

$L^2 = 36 + 64$

$L = 10$

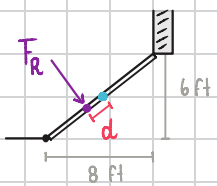


$A = 50\ ft^2$

$\theta \approx 36,87°$

$h_{cg} = (15 - 3)\ ft = 12\ ft$ · medido desde arriba

Centro de gravedad

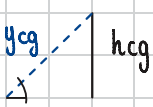


· Magnitud Fuerza resultante : $F_R = \gamma \cdot h_{cg} \cdot A$

$F_R = (64\ lbf/ft^3)(12\ ft)(50\ ft^2) = 38400\ lbf = 38,4\ Kips$

· Centro de presión :

$y_{cp} = \frac{I_{xc}}{y_{cg} \cdot A} + y_{cg}$



$y_{cg} = \frac{12}{0.6} = 20\ ft$

$I_{xc} = \frac{bh^3}{12} = \frac{5 \cdot (10)^3}{12} = 416.67\ ft^4$

↳ $y_{cp} = \frac{416.67}{20 \cdot 50} + 20 = 20,4167\ ft$

$d = y_{cp} - y_{cg} = 0,4167\ ft$

Fuerza de reacción en el punto A :

$\Sigma M_O = F_R \cdot (5 - d) - F_A \cdot 6 = 0$

$38400\ lbf \cdot 4,5833\ ft = F_A \cdot 6\ ft$

$F_A = 29333\ lbf$

Problema 2

Fuerza horizontal: $F_x = \gamma \cdot y_c \cdot A = \gamma_{H_2O} \cdot \left(\frac{h}{2}\right)(hL) = \gamma_{H_2O} \frac{h^2 L}{2}$

Fuerza vertical: $F_y = \gamma \cdot V = \int_0^L \int_0^b \int_0^{\beta x^{3/4}} \gamma_{H_2O} \, dy\,dx\,dz$

Tenemos que integrar porque y varía según x

$$= \gamma_{H_2O} \int_0^L \int_0^b (\beta x^{3/4}) \, dx\,dz$$

$$= \gamma_{H_2O} \int_0^L \left(\beta \cdot \frac{b^{7/4}}{7/4}\right) dz = \gamma_{H_2O} \beta \cdot \frac{b^{7/4}}{7/4} \cdot L$$

Entonces,

$$F_x = \gamma_{H_2O} \frac{h^2 L}{2}$$

$$F_y = \gamma_{H_2O} \beta \cdot \frac{b^{7/4}}{7/4} \cdot L$$

$$\|\vec{F}\| = \gamma_{H_2O} \sqrt{\left(\frac{h^2 L}{2}\right)^2 + \left(\gamma_{H_2O} \beta L \frac{b^{7/4}}{7/4}\right)^2}$$

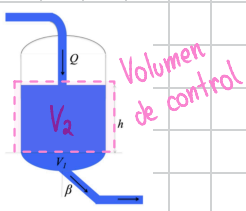
Posición de la fuerza:

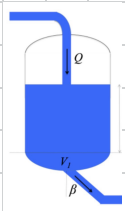
↳ y_p de F_x: $\frac{I_z}{y_c \cdot A} = \frac{(Lh^3/3)}{(h/2)(hL)} = \frac{2h}{3}$

↳ x_p de F_y: $\frac{\int_0^L \int_0^b \int_0^{\beta x^{3/4}} x \, dx\,dy\,dz}{\int_0^L \int_0^b \int_0^{\beta x^{3/4}} dx\,dy\,dz} = \frac{\beta \cdot \frac{4}{11} b^{11/4} \cdot L}{\beta \cdot \frac{4}{7} b^{7/4} \cdot L} = \frac{7}{11} b$

$(x_p, y_p) = \left(\frac{7b}{11}, \frac{2h}{3}\right)$

Problema 3 - Conservación de masa





$\} h(t)$

$$V = V_1 + V_2 = V_1 + h(t) \cdot \frac{\pi D^2}{4}$$

(V_1: cte; V_2: variable)

Ecuación de continuidad:

$$\frac{d}{dt}\left(\int_{V_c} \rho \, dV\right) + \int_A \rho \, \vec{u} \cdot \hat{n} \, dA = 0$$

$$\frac{d}{dt}(\rho V_c) - \rho \int_{A_{in}} \vec{u} \cdot dA + \rho \int_{A_{out}} \vec{u} \cdot dA = 0$$

$$\frac{d}{dt}\left(\rho\left(V_1 + h(t) \cdot \frac{\pi D^2}{4}\right)\right) - \rho Q_{in} + \rho \cdot Q_{out} = 0$$

$$\rho \frac{\pi D^2}{4} \frac{d}{dt} h(t) - \rho \cdot Q + \rho (v_1 \cdot A_1) = 0$$

$$\frac{d}{dt} h(t) = (Q - v_1 \cdot A_1) \frac{4}{\pi D^2}$$

$$\frac{dh}{dt} = \left(Q - v_1 \cdot \frac{\pi (0.02D)^2}{4}\right) \frac{4}{\pi D^2}$$

$$\boxed{\frac{dh}{dt} = \frac{4Q}{\pi D^2} - v_1 \cdot 0{,}002^2}$$